THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · HARD LEVEL CHAPTER TEST SET 2

Chapter 4: **Quadrilaterals** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The diagonals of a rectangle ABCD meet at O. If $\angle AOD = 110^\circ$, then $\angle OAB$ is: 1

(a) 35° **(b)** 55° **(c)** 70° **(d)** 110°

2. In an isosceles trapezium ABCD with AB $\parallel$ DC, $\angle A = (2x + 20)^\circ$ and $\angle D = (3x + 50)^\circ$. The measure of $\angle C$ is: 1

(a) 136° **(b)** 64° **(c)** 116° **(d)** 151°

3. The side of a rhombus is $(5x + 3)$ cm and its perimeter is 92 cm. The value of x is: 1

(a) 5 **(b)** 4 **(c)** 6 **(d)** 3

4. Which of these is NOT always true for a rhombus? 1

(a) All sides are equal **(b)** The diagonals are equal

(c) Opposite angles are equal **(d)** The diagonals bisect each other

5. The angles of a quadrilateral are in the ratio 4:5:7:8. The difference between the largest and the smallest angle is: 1

(a) 180° **(b)** 60° **(c)** 120° **(d)** 72°

6. One angle of a parallelogram is 4 times its adjacent angle. The smaller angle is: 1

(a) 36° **(b)** 45° **(c)** 56° **(d)** 144°

7. In a rhombus ABCD, $\angle ACD = 38^\circ$. Then $\angle ABC$ is: 1

(a) 52° **(b)** 104° **(c)** 142° **(d)** 76°

8. The diagonals of a rectangle ABCD meet at O. If $\angle AOD = 120^\circ$, then $\angle OAB$ is: 1

(a) 120° **(b)** 30° **(c)** 60° **(d)** 75°

9. **Assertion (A):** A parallelogram with one right angle is a rectangle. 1

Reason (R): A parallelogram with equal diagonals is a rhombus.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** The diagonals of a rhombus are equal. 1
Reason (R): The diagonals of a rhombus bisect each other at right angles.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. The angles of a quadrilateral are in the ratio 4:4:4:8. Find the largest angle and say whether it is acute, right or obtuse. 2

12. In a kite ABCD, AB = AD, CB = CD, $\angle A = 84^\circ$ and $\angle B = 96^\circ$. Find $\angle C$ and $\angle D$. 2

13. In a parallelogram ABCD, $\angle A = (6x + 3)^\circ$ and $\angle C = (7x - 2)^\circ$. Find x and $\angle B$. 2

14. The angles of a quadrilateral are in the ratio 5:5:6:8. Find the largest angle and say whether it is acute, right or obtuse. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. In a parallelogram ABCD with $\angle A = 84^\circ$, the bisectors of $\angle A$ and $\angle B$ meet at P. Find $\angle APB$. 3

16. The angles of a quadrilateral are x°, $(2x - 5)^\circ$, $(3x - 10)^\circ$ and $(4x - 15)^\circ$. Find all four angles. 3

17. One angle of a parallelogram is 4 times its adjacent angle. Find all the angles of the parallelogram. 3

OR

In a parallelogram ABCD with $\angle A = 96^\circ$, the bisectors of $\angle A$ and $\angle B$ meet at P. Find $\angle APB$. 3

18. The angles of a quadrilateral are x°, $(2x - 15)^\circ$, $(3x - 30)^\circ$ and $(4x - 45)^\circ$. Find all four angles. 3

OR

One angle of a parallelogram is 5 times its adjacent angle. Find all the angles of the parallelogram. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A rhombus-shaped tile has diagonals of length 16 cm and 12 cm. Another rhombus-shaped tile in the shop has one angle of 64°. 5
(i) At what angle do the diagonals meet?
(ii) Find the length of each side of the tile.
(iii) Find the perimeter.
(iv) Find the other three angles of the second tile and verify that the four angles add up to 360°.

20. In a quadrilateral ABCD, $\angle A : \angle B : \angle C : \angle D = 4 : 5 : 5 : 6$. Find the four angles. Show that AB $\parallel$ DC. 5

OR

In a parallelogram ABCD, AB $= (6x + 1)$ cm, CD $= (4x + 11)$ cm and AD = 14 cm. Find x, the lengths of AB and CD, and the perimeter. 5

END OF QUESTION PAPER — **Total: 40 Marks**